Diet, Lifestyle, and Acute Myeloid Leukemia in the NIH–AARP Cohort

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The relation between diet, lifestyle, and acute myeloid leukemia was assessed in a US cohort of 491,163 persons from the NIH–AARP Diet and Health Study (1995–2003). A total of 338 incident cases of acute myeloid leukemia were ascertained. Multivariate Cox models were utilized to estimate hazard ratios and 95% confidence intervals. Compared with those for never smokers, hazard ratios were 1.29 (95% confidence interval: 0.95, 1.75), 1.79 (95% confidence interval: 1.32, 2.42), 2.42 (95% confidence interval: 1.63, 3.57), and 2.29 (85% confidence interval: 1.38, 3.79) for former smokers who smoked ≤1 or >1 pack/day and for current smokers who smoked ≤1 or >1 pack/day, respectively. Higher meat intake was associated with an increased risk of acute myeloid leukemia (hazard ratio = 1.45, 95% confidence interval: 1.02, 2.07 for the fifth vs. first quintile; P for trend = 0.06); however, there were no clear effects of meat-cooking method or doneness level. Individuals who did not drink coffee appeared to have a higher risk of acute myeloid leukemia than those who drank various quantities of coffee. Neither fruit nor vegetable intake was associated with acute myeloid leukemia. This large prospective study identified smoking and meat intake as risk factors for acute myeloid leukemia.